Proficient in STAAD.Pro (80 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	**Overview of Structural Analysis and Design:** Basic Terminologies, Idealization of Structures, Understandingthe calculation of Shear force and Bending Moment **Introduction to STAAD.Pro:** Understanding the facilities in STAAD.Pro, Understanding the workflow ofSTAAD.Pro, Understanding the Types of Structure **STAAD.Pro Connect Edition GUI:** Understanding the Quick Access Toolbar, Ribbon Tab and Ribbon Toolbar,Understanding the Tool Search, Understanding the Workflow Page Control, Workflows Panel, Data Area andStatus Bar. **New File:** Creating a New File, Options in creating a new file, Opening an existing file using different methods. **Unit System:** Input Units, Changing the Graphical Display Units, Changing the Base Unit System, Convertingunits within a dialogue field and unit converter tool. **Coordinate system:** Understanding the Global Co-ordinate System, Understanding the Local Coordinatesystem, Understanding the relationship between Global and Local Coordinate System **STAAD Editor:** Starting the STAAD Editor, Understanding the STAAD Editor Interface, STAAD Input Files,Understanding the STAAD Command Language and Syntax, Editing Input Files	Theory/Demo	2.0 Hrs
2	Session 2	Exercise included for practice	Practical	2.0 Hrs
3	Session 3	**Introduction:** Understanding the methods of Creating Structure in STAAD.Pro, Understanding the StructureGeometry in STAAD.Pro **Grid Creation:** Understanding the Linear Grid Creation, Understanding the Radial Grid Creation, Understandingthe Irregular Grids, Importing STAAD.Pro Grid Files, Importing DXF Files as a Grid and Modifying the Grids usingthe Grid tool **Node Creation:** Creating the Nodes using Node Table, Creating the Nodes using Snap Node Option,Understanding the Node creation using STAAD Command Editor.	Theory/Demo	2.0 Hrs
4	Session 4	Exercise included for practice	Practical	2.0 Hrs
5	Session 5	**Beam Creation:** Understanding the Beam creation using Beam Layout, Creating Beams using Snap Beamoption, Understanding the beam creation with New Nodes, Creating Beams using Mid-points, Creating Beamsperpendicular to the existing beams, Understanding the Creation of Curved Beams and creation of beams byconnecting beams along X Axis/Y Axis/Z Axis and Understanding the Beam creation using STAAD Editor. **Structure Wizard:** Understanding the Structure Wizard Interface, Changing the units in Structure Wizard,Understanding various Model Types, Creating a Truss model, Creation of Frame Models, Creating ContinuousBeam, Creating Cylindrical Frame and beam model, Editing the Model's parameters, Importing CAD Models,Understanding the VBA Macro Models. Transferring the model to STAAD Pro Interface.	Theory/Demo	2.0 Hrs
6	Session 6	Exercise included for practice	Practical	2.0 Hrs
7	Session 7	**Selecting Objects in STAAD.Pro:** Understanding the Selection Group, Selecting Objects using Drag Box, DragLine and Drag Region, Understanding the Geometry Cursor, Understanding the All Geometry cursor andInverse Geometry Selection tools, Selecting Objects using Geometry List, Selecting Entities parallel to XY/XZ/YZPlane, Selecting Geometry Connecting to an entity , Understanding the Physical Members tool, Understandingthe Node Cursor tool, All Supports tools and Node List tool, Understanding the Beam Cursor tool, selectingbeams parallel to X/Y/Z axis, selecting beams connecting to a node/beam/plate/solid. Understanding theFiltered Selection, Selecting Objects by Property Name, Selecting objects that lack in input data. **Model Editing 1:** Inserting nodes in single member and in multiple members. Understanding the Break beamstool, stretching the selected beams, merging the selected beams, Understanding the renumber beams tool,Understanding the Intersect Beams tool. **Model Editing 2:** Creating copies using Translational Repeat tool along X Axis/Y Axis/Z Axis, Creating copiesusing Circular Repeat tool along the global axis, rotating the selected objects, Mirroring the selected objectsabout any plane, Moving entities **View Tools:** Understanding the Rotation Mode, 3D Rendering of Model, Controlling the labels, viewing theportion of the model, identifying the Beam Ends, Rotating the model to a predefined orientation. Exercises included for practice	Theory/Demo	2.0 Hrs
8	Session 8	Exercise included for practice	Practical	2.0 Hrs

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9	Session 9	**Section Profile Specification:** Creating a Prismatic Sections, Tee Section, Trapezoidal Section, Tapered Sections,Creating sections using User Provided Table option, Understanding the Section Database, Understanding thevarious Assignment Methods, **Material Specification:** Understanding the Isotropic Material Specification and Orthotropic MaterialSpecification, Understanding the Material constants **Node Specification:** Understanding the Node Specification Options, Rigid link between nodes **Member Specification:** Member Orientation, Axial Action Members, Member end release, Member EndOffsets, Member Imperfection, Cable Members, Cracked Section Properties, Member Fire Proofing, CreatingAttributes for New Beams. **Support Specification:** Fixed & Pinned Support, Roller Support, Spring Support, Multi-Linear Spring Support,Tension/Compression only Springs, Enforced Support, Inclined Support and Foundation Support **Group Specification:** Creating Group from Selection, Add Objects to an Existing Group. **Section Wizard:** Creating Irregular sections using section wizard & Section Database Manager	Theory/Demo	2.0 Hrs
10	Session 10	Exercise included for practice	Practical	2.0 Hrs
11	Session 11	**Loads:** Understanding the Load cases and Load Definitions, Creating Load Cases **Load Items:** Adding Self weight Load, Adding the Nodal Loads and Support Displacement, Understanding theMember Loads, Uniform Force and Moment, Concentrated Force and Moment, Linear Varying Load,Trapezoidal Load, Hydrostatic Load, Pre/Post Stress, Floor Load	Theory/Demo	2.0 Hrs
12	Session 12	Exercise included for practice	Practical	2.0 Hrs
13	Session 13	**Wind Load Generator:** Understanding the Wind Load Generator, Calculation of Design Wind Speed and DesignWind Pressure, Defining the wind load, Assigning the Wind Load **Load Combination:** Understanding the Load Combinations, Defining a new combination, Auto LoadCombination and Editing the Auto Combination Rules Analyzing the Model, Viewing Results in Output Viewer and Viewing Results in Post-Processing Mode, ViewingReactions, Displacements and Beam Results	Theory/Demo	2.0 Hrs
14	Session 14	Exercise included for practice	Practical	2.0 Hrs
15	Session 15	**Introduction to FEM:** Understanding the Finite Element Method, Plate and Shell Elements in STAAD.Pro **Plate Creation:** Creating Plates using Plate Layout, Snap Plate tool, creating 3 or 4-noded 2D plates usingexisting nodes, Creating Plates using infill plate tool, Creating Plates using STAAD Command Editor. **Structure Wizard:** Quadrilateral Plate, Cylindrical Surface, Hyperbolic Paraboloid Shell, Spherical Surface,Polygonal and circular plates with holes, Cooling Tower and Bunker or Silo. **Generate Meshing:** Generating Surface Mesh and Plate Mesh. **Parametric Modeling:** Understanding the Parametric Models, Creating parametric Surface Model, PolygonOpening / Circular Opening in a mesh model. Slab / Wall Connection	Theory/Demo	2.0 Hrs
16	Session 16	Exercise included for practice	Practical	2.0 Hrs
17	Session 17	**Solid Generation:** Creating 4 – 8 noded solids by connecting Existing nodes Solid 3D bricks from 4 to 8-noded,Solid Model Creation using Structure Wizard **Composite Decks:** Creation of Composite Decks **Selecting Plate Elements & Solid Elements:** Understanding the Plate Cursor and Solid Cursor **Plate Specification:** Thickness, Element Release	Theory/Demo	2.0 Hrs
18	Session 18	Exercise included for practice	Practical	2.0 Hrs
19	Session 19	**Plate Load:** Pressure on Full Plate, Concentrated Load, Partial Plate Pressure Load, Trapezoidal Load,Hydrostatic Load, Element Joint Load, Area Load, Solid Load, Temperature Load **Moving Load:** Understanding the Moving Loads, Defining a Vehicle for Loading, Generating Moving LoadCases, Adding Vehicles to the Load Generation **Load Envelope:** Understanding the Load Envelope and Defining the Load Envelope Reference Load & Repeat Load Viewing Plate Results	Theory/Demo	2.0 Hrs
20	Session 20	Exercise included for practice	Practical	2.0 Hrs
21	Session 21	**STAAD.Beava:** Creating Deck, Defining Roadway, Understanding the Vehicle Database, Defining the VehicleDatabase, Understanding the Load Generator Parameters, Defining the Load Generator Parameters, CreatingLoad Cases	Theory/ Demo	2.0 Hrs
22	Session 22	Exercise included for practice	Practical	2.0 Hrs

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23	Session 23	**Seismology:** Seismology Concept and basics of seismic waves, **General Principals for Earthquake Design:** Calculation of Design Horizontal Seismic Co-efficient, Calculation ofDesign Seismic Base Shear, Calculation of Distribution of Base shear to Diff floor levels **Seismic Definition in STAAD.Pro:** Understanding the parameters and generating the lateral seismic loads &Finding the Lateral Force by using STAAD.Pro	Theory/Demo	2.0 Hrs
24	Session 24	Exercise included for practice	Practical	2.0 Hrs
25	Session 25	**Push-over Analysis:** Types of Push-over analysis, Formation of Plastic Hinge, Building Performance **Dynamic Analysis:** Defining the Response Spectrum loads, Methodology of Response Spectrum Analysis,Understanding the Time History Analysis, Defining the Time History loads	Theory/Demo	2.0 Hrs
26	Session 26	Exercise included for practice	Practical	2.0 Hrs
27	Session 27	**Frequency Calculation:** Rayleigh Method & Modal Calculation Mass Modeling Using Reference Loads **P-Delta Analysis:** Understanding the P-Delta Effect, Types of P-Delta Effects, Methodology and performing theP-Delta Analysis **Buckling Analysis:** Understanding the buckling analysis, performing Buckling Analysis and Understanding theImperfection Analysis **Post-Processing Workflow:** Understanding the workflow, creating animated videos from results, viewingvarious results using Post-Processer and Output Viewer.	Theory/Demo	2.0 Hrs
28	Session 28	Exercise included for practice	Practical	2.0 Hrs
29	Session 29	**Concrete Design in STAAD.Pro:** Understanding the Beam Design and Column Design, Understanding theDesign Parameters, Specifying Concrete Design Parameters, Specifying Design Commands, Discussing theDesign Results	Theory/Demo	2.0 Hrs
30	Session 30	Exercise included for practice	Practical	2.0 Hrs
31	Session 31	**Interactive concrete design:** Understanding the RC Designer Interface, Design Overview in RC Designer,Creating Physical Members, Creating Design Briefs for and Design Groups for beams and columns, DesignCheck	Theory/Demo	2.0 Hrs
32	Session 32	Exercise included for practice	Practical	2.0 Hrs
33	Session 33	**Steel Design in STAAD.Pro:** Design Process, Understanding the Steel Design in STAAD.Pro, Specifying SteelDesign, Understanding the Steel Design Parameters and specifying the steel design parameters, specifyingsteel Design Commands, Discussing the Deign Results Design of Overhead Transmission Line Towers	Theory/Demo	2.0 Hrs
34	Session 34	Exercise included for practice	Practical	2.0 Hrs
35	Session 35	**Foundation Design:** Types of Foundations, Foundation design in STAAD Foundation Advanced, understandingthe workflow of STAAD Foundation **Isolated Footing:** Creating New File, Defining Project Info, Specifying the Column Position, Defining the Loads,Load Combinations, Specifying Design Parameters and Design Discussion **Combined / Strip Footing:** Understanding the Design parameters and Design Discussion	Theory/Demo	2.0 Hrs
36	Session 36	Exercise included for practice	Practical	2.0 Hrs
37	Session 37	**Mat Foundation:** Specifying Mat Boundary, Creating Mesh, Analyzing the Slab, Moment Envelope Generation,Designing the Slab, Reinforcement Zones **Pile Cap Design:** Understanding the Design Parameters and design discussion **Plant Foundation Design:** Vertical Vessel Foundation, Horizontal Vessel Foundation, Tank Foundation **Tool Kit:** Ribbed Beam Footing	Theory/Demo	2.0 Hrs
38	Session 38	Exercise included for practice	Practical	2.0 Hrs
39	Session 39	Importing Various Models **Export:** Export to DXF, Export to CIS/2 File, Export to SACS Input File **Plotting from STAAD.Pro:** Plot using the Print Current View tool, Take Picture Tool **Report Setup:** Report Page, Adding Items, , Adding Picture, Adding Load Cases, Adding Name and Logo	Theory/Demo	2.0 Hrs
40	Session 40	Exercise included for practice	Practical	2.0 Hrs